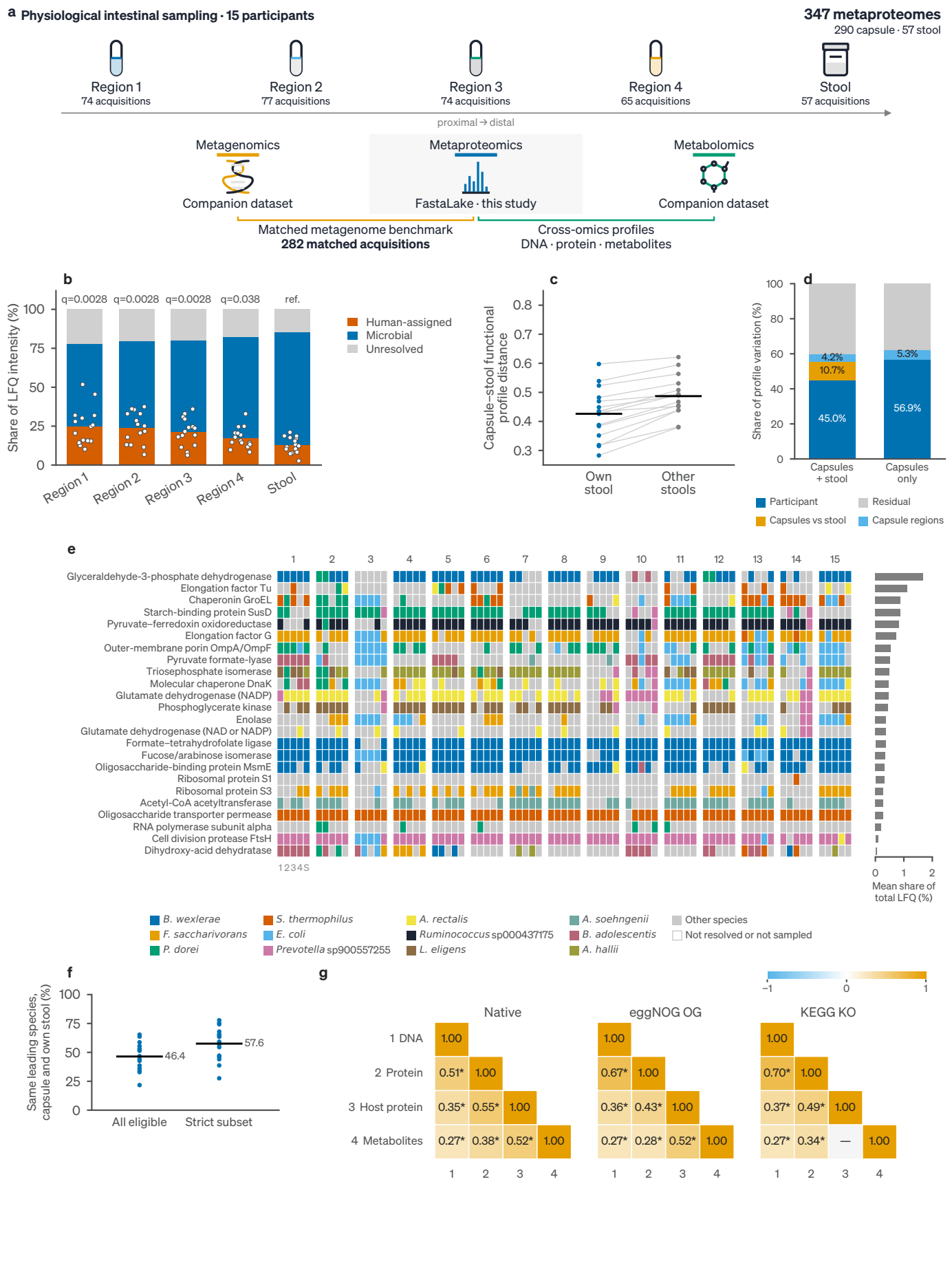

a
Physiological intestinal sampling · 15 participants
347 metaproteomes
290 capsule · 57 stool
Region 1
74 acquisitions
Region 2
77 acquisitions
Region 3
74 acquisitions
Region 4
65 acquisitions
Stool
57 acquisitions
proximal → distal
Metagenomics
Companion dataset
Metaproteomics
FastaLake · this study
Metabolomics
Companion dataset
Matched metagenome benchmark
282 matched acquisitions
Cross-omics profiles
DNA · protein · metabolites
b
q=0.0028
q=0.0028
q=0.0028
q=0.038
ref.
Share of LFQ intensity (%)
Region 1
Region 2
Region 3
Region 4
Stool
Human-assigned
Microbial
Unresolved
c
Capsule–stool functional profile distance
Own stool
Other stools
d
Share of profile variation (%)
4.2%
10.7%
45.0%
5.3%
56.9%
Capsules + stool
Capsules only
Participant
Residual
Capsules vs stool
Capsule regions
e
Glyceraldehyde-3-phosphate dehydrogenase
Elongation factor Tu
Chaperonin GroEL
Starch-binding protein SusD
Pyruvate–ferredoxin oxidoreductase
Elongation factor G
Outer-membrane porin OmpA/OmpF
Pyruvate formate-lyase
Triosephosphate isomerase
Molecular chaperone DnaK
Glutamate dehydrogenase (NADP)
Phosphoglycerate kinase
Enolase
Glutamate dehydrogenase (NAD or NADP)
Formate–tetrahydrofolate ligase
Fucose/arabinose isomerase
Oligosaccharide-binding protein MsmE
Ribosomal protein S1
Ribosomal protein S3
Acetyl-CoA acetyltransferase
Oligosaccharide transporter permease
RNA polymerase subunit alpha
Cell division protease FtsH
Dihydroxy-acid dehydratase
1234S
Mean share of total LFQ (%)
B. wexlerae
F. saccharivorans
P. dorei
S. thermophilus
E. coli
Prevotella sp900557255
A. rectalis
Ruminococcus sp000437175
L. eligens
A. soehngenii
B. adolescentis
A. hallii
Other species
Not resolved or not sampled
f
Same leading species, capsule and own stool (%)
46.4
57.6
All eligible
Strict subset
g
Native
eggNOG OG
KEGG KO
1 DNA
2 Protein
3 Host protein
4 Metabolites
1.00
0.51* 1.00
0.35* 0.55* 1.00
0.27* 0.38* 0.52* 1.00
0.67* 1.00
0.36* 0.43* 1.00
0.27* 0.28* 0.52* 1.00
0.70* 1.00
0.37* 0.49* 1.00
0.27* 0.34* — 1.00